A PROPER CURVE WITH H^0(O_X) = K IS CONNECTED, COHEN-MACAULAY AND EQUIDIMENSIONAL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument:

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Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The maximal submodule supported in closed points. Divisors, Lemma \label{divisors-lemma-remove-embedded-points}. Proof lines 1289-1297: the maximal coherent submodule I of O_X supported in closed points exists, and its vanishing is equivalent to X having no embedded points.
- Sheaves with zero-dimensional support are globally generated. Varieties, Lemma \label{varieties-lemma-chi-tensor-finite}. Proof lines 1291-1293: I is globally generated; since 1 in Gamma(X, O_X) is not a section of I, one gets I = 0.
- No embedded points gives Serre's S1. Divisors, Lemma \label{divisors-lemma-S1-no-embedded}. Proof lines 1297-1299: X has (S_1) because it has no embedded points, hence is Cohen-Macaulay in dimension 1.

Source reference. Original lemma 0346: sources/stacks-chow/tags.txt, source lines 4081–4081. Prerequisite pointer invoked at proof line(s) 1299; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02OL: sources/stacks-chow/tags.txt, source lines 3520–3520. Prerequisite pointer invoked at proof line(s) 1291; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AYT: sources/stacks-chow/tags.txt, source lines 13801–13801. Prerequisite pointer invoked at proof line(s) 1293; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k having dimension* 1 *and $H^0(X, \mathcal{O}_X) = k$. Then X is connected, Cohen-Macaulay, and equidimensional of dimension* 1.

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Proof. Since $\Gamma(X, \mathcal{O}_X) = k$ has no nontrivial idempotents, we see that X is connected. This already shows that X is equidimensional of dimension 1 (any irreducible component of dimension 0 would be a connected component). Let $\mathcal{I} \subset \mathcal{O}_X$ be the maximal coherent submodule supported in closed points. Then $\mathcal{I}$ exists (Divisors, Lemma 02OL) and is globally generated (Varieties, Lemma 0AYT). Since $1 \in \Gamma(X, \mathcal{O}_X)$ is not a section of $\mathcal{I}$ we conclude that $\mathcal{I} = 0$. Thus X does not have embedded points (Divisors, Lemma 02OL). Thus X has (S_1) by Divisors, Lemma 0346. Hence X is Cohen-Macaulay. $\square$